

Total Duration : 1 hour 20 minutes

Total Marks: 50 (will be converted to 25).

CO Distribution:

CO1 - Answer any one out of two questions. There might be several sub-sections of each question such as 1a.(i),(ii) and so on. The questions will test your theoretical knowledge and concepts. It carries 10% of total marks [5 marks]. Spend the least amount of time for this section.[10 mins at max]

CO2 - Answer any one out of two questions. There might be several sub-sections of each question such as 2a.(i),(ii) and so on. The questions will be a combination of derivations and theories both so it is essential both. It carries 30% of total marks. [15 marks]

CO3 - Answer any two out of three questions. There might be several sub-sections of each question such as 3a.(i),(ii), 3b(i),(ii) and so on. The questions will be purely based on maths and the structure is quite similar to what you have seen in quizzes, assignments, class examples. It carries 60% of total marks. [30 marks]

Common mistake: Answering all the questions from CO1,CO2, and CO3. Kindly keep this in mind before starting the exam that there will be options to choose from.

Tip for Exam:

Start with CO3/CO2 and then answer CO1 at last, this helps with efficient time management.

Note that skipping chapters is not an option since the questions are not made purely based on one chapter it is always a combination of multiple concepts.

Chapter - 01

- Floating Point Number Representation for Conventions 01,02 and 03. [both negative and non-negative support]
- IEEE Standard Format for double precision.
- Possible number of combinations of numbers for Convention 01, 02 and 03.
- Rounding Error Maths using number line. Try to use the manual process that I have shown in class for conversion since calculator conversions might have decimal value inconsistency.
- Scale Invariant Rounding Error formula: if there is any difference between $\text{fl}(x*y)$ and $(x*y)$ which is usually always the case.
- The maximum value of Scale Invariant Rounding Error or delta max is known as Machine Epsilon.
- Derivation - We get the formulas for Machine Epsilon for Convention 01, 02 and 03 from the Scale Invariant Rounding Error Formula. See the derivations given using examples both in my lecture notes and chapterwise notes.
- Concept of overflow and underflow.
- What is loss of significance? How do we resolve it?

Chapter-02

- Representation of polynomial - relationship between degree(n), no of nodes(n+1) and coefficients(n+1).
- Taylor Series - General Representation
- Vandermonde Matrix - representation, disadvantage and calculator tricks for finding determinant and inverse matrix[given in discord pinned message].
- Lagrange Interpolation method - representation[using Lagrange Basis], better than Vandermonde, disadvantage.

- **Newton's Divided Difference - representation**[using divided differences - tree structure while solving maths], better than Lagrange, no disadvantage.
- **Hermite Interpolation - representation**[using Hermite Basis], advantage[degree increased without increasing no of nodes due to addition of derivative], no disadvantage.
- **Cauchy's Theorem** used for finding upper bound of Interpolation Error, maximize the functions separately, if the range for z_i function is not given then the values for given x /nodes can be considered the range for z_i function.
- **Runge Phenomenon** and it's solutions [Piecewise interpolation and Chebyshev Nodes=equally spaced angles instead of equally spaced points/nodes]. The formula for Chebyshev nodes is different for symmetric and asymmetric interval. You may go through my class recording both cases have been discussed.

Chapter-03

- **Forward Difference, Backward Difference, Central Difference** [gives approximate value of Derivative]
- **Truncation Error, |Actual derivate - Central/Forward/Backward Difference|**
- **Upper Bound of Truncation Error**, If the range of z_i is not given, you need to find the range for z_i using the following formulas:
 - Central Difference: $[x-h, x+h]$,
 - Forward Diference: $[x, x+h]$,
 - Backward Diference: $[x-h, x]$
- **Rounding Error and Truncation Error relationship with step size(h).**
- **Relative Error:**

$$\frac{|\text{actual derivative} - \text{approximate derivative}|}{|\text{actual derivative}|}$$

For percentage relative error, you need to multiply by 100.

- Richardson Extrapolation [gives better result since it omits the h^2 term from error].
- How do we get the formula for Richardson Extrapolation[class example discussed using step size $h/2$, Assignment-03 given using step size $4h/3$]
- Examples related to Richardson Extrapolation [get familiar with the notations used in class examples and assignments quizzes]
- If you face Richardson Extrapolation in CO2 you need to show derivation, if you face Richardson Extrapolation in CO3 you need to solve math using the formula directly[you do not need to show derivation here].

Break Down of the Midterm Syllabus according to book chapters:

✓ Chapter 1:

Floating Point Arithmetic. (Ch:1.1-1.3)

Rounding error, loss of significance. (Ch:1.4-1.5)

✓ Chapter 2:

Polynomial Interpolation, Taylor Series, Vandermonde method. (Ch:2.1-2.2)

Lagrange form. (Ch:2.3).

Newton divided/difference form. (Ch:2.4).

Interpolation error, Cauchy's Theorem. (Ch:2.5)

Runge's Phenomenon and Chebyshev Nodes, Hermite Interpolation method. (Ch:2.6-2.7)

✓ Chapter 3:

Differentiation: higher order finite difference [FD, BD, CD]. (Ch:3.1.)

Rounding error, Richardson extrapolation. (Ch:3.2-3.3)

Resources to Study from:

Rafeed Sir's Playlist:

<https://docs.google.com/document/d/1iBB0qw32X-3nGlKcYcpWWIX0X0gQtUf4DEGzCzC9CLs/edit?tab=t.0>

Class Lecture Notes[Concised Notes & Chapterwise Notes]:

https://drive.google.com/drive/folders/1zNPhDIJksxd-mrqr5Qe_mYXpLN_TgRQ?usp=sharing

Recordings of this semester's online classes and some previous recordings as well:

https://www.youtube.com/playlist?list=PLKHc9HBF0yJxhkfBHnZt0ZRs8d3_ghcYq

Playlist link of AMK sir and Aquib Sir:

 Playlist Links